

Pandas + GroupBy + Join — Student Practice Task

Questions only. No trainer challenge section.

Part A — Single Dataset: Orders

1. Load `orders_dataset.csv` into a DataFrame and display the first 5 rows.
2. Display the last 5 rows and the shape of the DataFrame.
3. Display column names, data types, and basic information about the dataset.
4. Find the number of unique products and unique cities.
5. Display all orders where Category is Electronics.
6. Display orders where Quantity is greater than 2.
7. Find the number of orders for each Status using `value_counts()`.
8. Calculate the total revenue of every order using Quantity and UnitPrice.
9. Find the total revenue generated from all orders.
10. Find the average UnitPrice for each Category using `groupby()`.
11. Find the total Quantity sold for each Product Category.
12. Find the total revenue generated from each City.
13. Find the average order value for each Status.
14. Find the maximum TotalAmount for each Category.
15. Sort the DataFrame by TotalAmount from highest to lowest.
16. Display the top 5 highest-value orders.
17. Find the product with the highest TotalAmount.
18. Count how many orders were placed in each City and Status combination using `groupby()`.
19. Find the total revenue from Completed orders only.
20. Find cities whose total revenue is greater than 20000 using `groupby()` and a condition.
21. Create a new column PriceLevel: 'High' if UnitPrice ≥ 10000 , otherwise 'Normal'.
22. Create a new column OrderValueLevel: 'Large' if TotalAmount ≥ 15000 , otherwise 'Regular'.
23. Find the average Quantity for Electronics and Furniture separately.
24. Find the category with the highest total revenue.
25. Use `groupby()` with multiple aggregations to calculate total quantity, average unit price, and total revenue for each Category.

Part B — Joining Orders and Customers

26. Load `customers_dataset.csv` into a second DataFrame.

27. Perform an inner join between orders and customers using CustomerID.
28. Select and display CustomerName, Product, Quantity, TotalAmount, Membership, and City after the join.
29. Find the total revenue generated by each customer.
30. Find the total revenue generated by each Membership type.
31. Find the average order value for each Membership type.
32. Find the total quantity purchased by customers from each City.
33. Find the number of orders made by each Gender.
34. Find the average customer age for each Membership type.
35. Find the customer who generated the highest total revenue.

Part C — Conditions + Join

36. After joining the datasets, display only Gold members who placed Completed orders.
37. Find Gold members whose TotalAmount is greater than 10000.
38. Find customers older than 30 who placed Completed orders.
39. Find Female customers from Kochi and display their order details.
40. Find Male customers who purchased Electronics products.
41. Find Silver members whose Quantity is greater than 1.
42. Find customers from Thrissur whose orders have TotalAmount greater than 5000.
43. Find Gold members whose total spending is greater than 15000 using groupby() and a condition.
44. Find the total revenue of Completed orders for each Membership type.
45. Find the total revenue generated by each City for Gold members only.
46. Find the average TotalAmount for Male and Female customers separately.
47. Find customers who have at least one Cancelled order.
48. Find customers who purchased Furniture and have Gold membership.
49. Find the top 3 customers based on total Completed-order revenue.
50. Create a final customer summary containing CustomerName, Membership, City, number of orders, total quantity purchased, and total revenue.